

CHRISTIAN AMOS OTIENO

Phone +254 713114123 | [E-mail](#) | [Linkedin](#) | [Github](#)

PROFESSIONAL SUMMARY

Adaptive, highly analytical, and results-driven **Full-Stack Software Engineer** with extensive experience architecting scalable backend systems, high-performance web applications, and database integrations. Specialized in leveraging **Go (Golang)** for high-concurrency services, network protocols, and core computer science optimization tasks, alongside modern frontends using **TypeScript**, **Next.js**, and **Tailwind CSS**. Adept at full lifecycle software engineering, containerization with **Docker**, database modeling with **PostgreSQL/ Prisma**, and implementing secure authentication workflows.

TECHNICAL SKILLS

- **Languages:** Go (Golang), TypeScript, JavaScript (ES6+), Python, SQL, HTML5, CSS3
- **Frameworks & Libraries:** Next.js, React, Node.js, Prisma ORM, NextAuth, Tailwind CSS
- **Databases & Caching:** PostgreSQL, Redis, Database Schema Design & Migration
- **Core CS Fundamentals:** Data Structures & Algorithms, Network Programming (TCP/UDP), Asynchronous I/O, Dynamic Programming, Concurrency Models, RESTful APIs
- **DevOps & Tools:** Docker, Git, GitHub, Linux, Bash Scripting, Render, Webpack, Postman

KEY PROJECTS

Hyperlocal Gift Economy | Full-Stack Sharing Platform [Github](#)

Technologies: Next.js, TypeScript, Docker, Prisma ORM, PostgreSQL, Tailwind CSS

- Architected and deployed a full-stack hyper-local gifting platform enabling community members to post, request, and exchange items seamlessly.
- Containerized application microservices using **Docker** to ensure smooth staging, multi-environment consistency, and streamlined deployment pipelines.
- Designed type-safe PostgreSQL relational schemas using **Prisma ORM**, optimizing database queries for high transaction throughput.

Airbnb Booking System Backend | High-Concurrency Reservation Engine

Technologies: Go (Golang), Redis, PostgreSQL, RESTful API

- Engineered a robust backend reservation system handling simultaneous booking requests with sub-millisecond response times.
- Implemented temporary distributed locks using **Redis** to prevent double-booking race conditions during high-concurrency peak traffic.

Core Systems & Algorithmic Projects — Go (Golang) Suite

- **Groupie-Tracker:** Built a data-driven web portal consuming complex REST APIs, implementing dynamic data visualization and Go backend server handlers.
- **Tetris-Optimizer:** Developed a spatial layout optimization algorithm using recursive backtracking to fit irregular shapes in minimal square grids.
- **Push-Swap:** Designed highly optimized stack sorting algorithms in Go, calculating minimal instruction sequences under strict operational constraints.
- **Net-Cat:** Programmed a multi-client TCP chat server in Go featuring thread-safe channel broadcasting, concurrent connection

handling, and message logging.

PROFESSIONAL EXPERIENCE

Apprentice Software Engineer — Zone01 Kisumu 2026 – Present •

Participating in an intensive project-based software engineering apprenticeship specializing in full-stack architecture and low-level system design.

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Built 8+ computer science projects adhering to peer-reviewed code quality, strict testing protocols, and clean architecture principles. Engineered custom HTTP web servers, CLI tools, and concurrency models in native Go without external framework dependencies. •

EDUCATION & SPECIALIZED CREDENTIALS

Neuro-Analytics & Brain-Data Integration — Skills for Africa 2023 – 2024 **Bachelor of Science in Microbiology and Biotechnology** — Aga Khan University 2019 – 2022